

Three New Methods to Find Initial Basic Feasible Solution of Transportation Problems

Eghbal Hosseini

Department of Mathematics
University of Raparin, Ranya, Kurdistan Region Iraq

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Abstract

The transportation problems have attracted many researchers in optimization because of their applications in several areas of science and real life. Therefore, solving of these problems especially finding initial basic feasible solution for them would be significant. Although there are some heuristic approaches to find initial solution, but there is no any efficient algorithm with its Matlab code to solve random large size transportation problems. In this paper, an efficient algorithm in three cases with its Matlab code is proposed which even is better than the best algorithm in references by solving some test problems and also our generated large size problems. The algorithm is forcefully efficient for problems with large size and it has sufficiently suitable results, at least in on case of the algorithm, for test problems in the references according to computational results.

Keywords: Transportation problems; heuristic approaches; random large size; initial basic feasible solution

1 Introduction

The transportation problem is one of the most important problem in different science. This problem transports some products from sources to destinations with minimum cost of transportation and satisfaction of the demand and the supply constraints. One of the significant form of linear programming problem is transportation which can be expanded to inventory, assignment, traffic and so on. For analyzing and formulation of some model, transportation problems are required [1-3]. Therefore solving transportation problem, finding minimal total cost, would be remarkable [4-8]. Proposing optimal solution needs to start from a feasible solution

as an initial basic feasible (IBFS) solution. Therefore initial feasible solution is basic solution which affects to optimal solution for the problem. In other words, finding IBFS would be significant. Although several meta-heuristic approaches have been proposed to solve optimization problems [9-12], but there just few methods in references which can find IBFS for the problem [13-15]. These methods are as follows: Northwest Corner Method (NCM), this method begins from the northeast cell in the transportation table. The algorithm allocates the possible maximum amount to the cell and regulates supply and demand quantities. Then each supply or demand which attains zero, correspond row or column will exit from the rest of calculations. This process will be continued until just a row or column is left from the table. During all these calculations the northwest cell of the table, without regarding removed rows and columns, will be selected each time. Minimum Cost Method (MCM), in this method, a cell which has lower cost is selected sooner than a cell with higher cost. In fact, the appropriation starts from the cell which includes the least cost of the table. The Minimum Cost Method finds IBFS better than NCM because the algorithm regards costs during the allocation unlike NCM. All process in NCM will be repeated here just with this difference that always the cell which includes minimum cost is selected instead the northwest cell. Minimum Row Method, Vogel's Approximation Method (VAM), this method has the best results according to literature. Summary of VAM steps are as follows: at the first, the difference between two least costs is calculated for each row and column as a penalty. Then a row or column which has the biggest penalty will be chosen. The possible maximum amount is allocated to the cell with the minimum cost in the chosen row or column. Each row which has no supply or the column which the demand has been totally satisfied for that will be removed from the rest of calculations. The difference between two least costs will be calculated for the remaining rows and columns and the process is continued to find an initial basic feasible solution. Rahul Method, this method has high computing in each iterations. Rahul Method chooses the cell with the largest negative value of w_{ij} in each iteration which $w_{ij} = c_{ij} - u_i - v_j$. u_i is the value of the biggest cost in row i and v_j is the value of the biggest cost in column j . Removing of rows and columns and also the amount allocated are similar to the previous methods.

In generally, the best initial basic feasible solution is found by Vogel's Approximation Method and the worst IBFS is generated by Northwest Corner Method. Meanwhile the least number of calculations is related to Northwest Corner Method. In all methods, the supply and demand must be equal. So if supply is more than demand, then a dummy column will be added in the table of transportation. Costs of cells in this dummy column should be zero and its demand is equal to the difference between supply and demand at the first time. As well as, if demand is more than supply, then a dummy row will be added in the table of transportation. Costs of cells in the row is zero and its supply is equal to the difference between supply and demand at the first time.

However there are heuristic approaches to find initial feasible solution, but there is no any attempt for proposing general code of algorithms in Matlab for example. In this paper, the authors has tried to propose three new algorithms to find initial feasible solution and also to propose Matlab codes of Vogel method and the best proposed algorithm in the paper. Therefore many more problems can be solved in the future by these codes.

Majority of previous proposed algorithms have been used to solve small or specific problems. In fact, large size and random problems never have unresolved in literature. In this paper, Matlab codes have been proposed to generate different problems in large or small sizes and comparison of the some algorithms and our algorithm has been presented by these generated problems.

The best results are related to Vogel algorithm in references therefore for all examples, proposed algorithms have been compared with Vogel algorithm and one of our algorithm is much better than Vogel almost for all examples. Briefly this paper has three novelty: three new algorithm to find initial solution of transportation, Matlab codes for proposed algorithms and Vogel approximation and Matlab codes to generate transportation problems in different sizes.

2 Total Differences Method 1(TDM1)

In this section three new algorithms will be discussed in details and step by step. TDM1, TDM2 and TDSM have been constructed from scrutiny of relationship of costs in rows and columns. Therefore in this section the main concepts of the proposed algorithms and their steps to solve transportation problems are discussed.

2.1 Main Step of TDM1

Vogel algorithm uses penalties for all rows and columns, TDM1 calculates penalties only for rows. The algorithm uses following formula to calculate these penalties:

$$\alpha_i = \sum_j (c_{ij} - c_r) \quad (1)$$

Which $c_r = \min_j \{c_{ij}\}$ & i is the number of the row}. In fact all differences between each cost and minimum cost will be determined. In Vogel algorithm, two least elements are important for each row and column. But all costs affect in a feasible solution of transportation problem, so TDM1 uses all costs for each row instead. To calculate penalties of each row by TDM1 just a simple code in Matlab is required. The pseudo code of that is proposed in Table 1.

Table 1 - Pseudo code of penalties of rows by TDM1

begin
Number of rows is m
Number of column is n
for(i=1 to m)
Sort vector of costs in ith row (c(i))
sum=0;
for(j=2 to n)
sum=sum+(c(1,j)-c(1,1));
end
end
end

2.2 Main Step of TDSM

The method of least squares is an appropriate method to define error, TDSM uses this idea to determine total differences between minimum cost and the other. The algorithm uses following formula to calculate these penalties:

$$\alpha_i = \sum_j (c_{ij} - c_r)^\theta \quad (2)$$

Which $c_r = \min_j \{c_{ij} \& i \text{ is the number of the row}\}$ and $1 < \theta \leq 2$. In fact, using $\theta > 1$ causes to clarify the differences between costs and the least cost. The pseudo code of this step in MATLAB has been shown in Table 2.

Table 2 - Pseudo code of penalties of rows by TDSM

begin
Number of rows is m
Number of column is
$1 < \theta \leq 2$
for(i=1 to m)
Sort vector of costs in ith row (c(i))
sum=0;
for(j=2 to n)
sum=sum+(c(1,j)-c(1,1)). ^θ ;
end
end
end

2.3 Main Step of TDM2

TDM1 calculates penalties only for rows, TDM2 uses (1) for all rows and columns. By calculating penalties for rows and columns, there will be more chooses and so maybe better feasible solution will be gained. All steps of TDM1 to propose an initial feasible solution are described as follows:

1. If the total supply and demand aren't equal, make them equal by adding a dummy row or a dummy column.

2. For each row of the transportation table, find the total differences between the least and each other costs. In fact the algorithm corresponds α_i for i th row according to the following formula:

$$\alpha_i = \sum_j (c_{ij} - c_r)$$

Which $c_r = \min_j \{c_{ij} \mid i \text{ is the number of the row}\}$.

3. Choice a row with the greatest total differences.
4. The highest possible amount will be assigned to the minimum cost cell of the chosen row to satisfy demand.
5. A row or a column which has no supply quantity available or the demand is completely satisfied will be removed.
6. Using (1), total differences of the costs in rows without regarding the removed row/rows and column/columns will be calculated.
7. While an initial feasible solution hasn't been found go back to the step 3.

All above Steps in two other algorithms are same except Step 2. This Step for TDSM and TDM2 algorithms have been proposed in section 3.2 and 3.3 respectively.

2.5 Feasibility and efficiency

Proposed algorithm in three cases has been run for several times for solving different sizes of transportation problems. The algorithm, in all three cases, is feasible for small size of the problems based on Examples 1-3. Also the algorithm is completely feasible for moderate size of the problems but it is efficient just by TDM1 according to Examples 4-9. Calculations of TDM1 is less in comparison with other methods, because only rows are assigned by penalties. Also in small problems TDM1 has less choices than Vogel algorithm which allocates all of rows and columns by penalties. So the algorithm should be more efficient for larger size of problems. This claim has been confirmed according to the computational results in Examples 10 -15.

3 Computational results

Example 1

Consider the following transportation problem:

	D1	D2	D3	D4	Supply
O1	20	22	17	4	120
O2	24	37	9	7	70
O3	32	37	20	15	50
Demand	60	40	30	110	240

Step 1: Total supply and demand are equal, therefore dummy row or column aren't need.

Step 2: Using (1), the total differences between the minimum cost and each other costs for each row is calculated.

Total Differences

O1 47
O2 49
O3 44

Step 3: Second row will be selected as the greatest total differences.

Step 4: The minimum cost cell of the second row is 7 and the highest possible amount for that is 70. Table of transportation problem will be changed to the following Table.

	D1	D2	D3	D4	Supply	
O1	20	22	17	4	120	
O2	24	37	9	7	70	0
O3	32	37	20	15	50	
Demand	60	40	30	40	240	

Step 5: According to the recent Table the second row has no supply quantity, so it will be removed from the rest process of the algorithm.

Step 6: Using (1), total differences of the costs in rows without regarding the removed second row is calculated.

Total Differences

O1 47
O2 ---
O3 44

Step 7: By going back to the step 3 the first row is selected. (Iteration 2):

Step 4: The minimum cost cell of the first row is 4 and the highest possible amount for that is 40.

	D1	D2	D3	D4	Supply	
O1	20	22	17	4	40	80
O2	24	37	9	7	70	0
O3	32	37	20	15	50	
Demand	60	40	30	0	240	

Step 5: The fourth column is removed because its demand has been completely satisfied.

Step 6: Using (1), total differences of the costs in rows without regarding the removed second row and fourth column is calculated.

Total Differences

O1 5
O2 ---
O3 29

Step 7: By going back to the step 3 the third row is selected. (Iteration 3):

Step 4: The minimum cost cell of the third row is 20 and the highest possible amount for that is 30

	D1	D2	D3	D4	Supply	
O1	20	22	17	4	40	80
O2	24	37	9	7	70	0

O3	32	37	20	30	15	20
Demand	60	40	0	0	240	

Step 5: The third column is removed too.

Step 6: Using (1),

Total Differences

O1	2
O2	---
O3	5

Finally, the Table which proposes a feasible solution is obtained: D1 D2

	D3	D4	Supply				
O1	20	40	22	40	17	4	40
O2	24	37	9	7	70	0	
O3	32	20	37	20	30	15	0
Demand	0	0	0	0	240		

Cost of proposed feasible solution is: $Cost = 20 \times 40 + 22 \times 40 + 4 \times 40 + 7 \times 70 + 32 \times 20 + 20 \times 30 = 3570$

Example 2

Consider the following transportation problem:

	D1	D2	D3	D4	Supply
O1	3	5	7	6	50
O2	2	5	8	2	75
O3	3	6	9	2	25
Demand	20	20	50	60	150

Using the proposed algorithm 1, feasible solution is found according to the following table.

	D1	D2	D3	D4	Supply		
O1	3	5	20	7	30	6	50
O2	2	20	5	8	20	2	35
O3	3	6	9	2	25	25	
Demand	20	20	50	60	150		

Correspond cost of the Table is: $Cost = 5 \times 20 + 7 \times 30 + 2 \times 20 + 8 \times 20 + 2 \times 35 + 2 \times 25 = 630$

Example 3

Consider the following transportation problem:

	D1	D2	D3	D4	Supply
O1	19	30	50	10	7
O2	70	30	40	60	9
O3	40	8	70	20	18
Demand	5	8	7	14	34

A feasible solution will be found using the proposed algorithm 1 according to the following table.

	D1	D2	D3	D4	Supply		
O1	19	5	30	50	10	2	7

O2	70	30	40	7	60	2	9
O3	40	8	8	70	20	10	18
Demand	5	8	7	14	34		

Cost of the solution is: $Cost = 19 \times 5 + 10 \times 2 + 40 \times 7 + 60 \times 2 + 8 \times 8 + 20 \times 10 = 779$

A comparison among the first and second algorithms in this paper and other methods has been shown in Table 5 for Examples 1-3. Algorithm of Vogel has the best results in references, both TDM 1 and TDSM have better solution than Vogel's solution in Example 2. Also TDM 1 works same as Vogel algorithm for Example 3. Improvement amount by TDM 1 and TDSM have been shown in Tables 6 and 7.

Table 3 – Comparison among TDM1, TDSM and other algorithms

	North-West	Min-Cost	Vogel	TDM 1	TDSM
Example 1	3680	3670	3520	3570	3710
Example 2	670	650	650	630	610
Example 3	1015	814	779	779	781

Table 4 – Improvement amount of Vogel algorithm by TDM1

	Vogel	TDM 1	Improvement
Example 1	3520	3570	-0.01%
Example 2	650	630	0.3%
Example 3	779	779	0%

Table 5 – Improvement amount of North-West and Min-Cost algorithms by TDSM

	North-West	Min-Cost	TDSM	Improvement1	Improvement2
Example 1	3680	3670	3710	-0.008%	-0.01%
Example 2	670	650	610	0.09%	0.06%
Example 3	1015	814	781	0.23%	0.04%

To prove efficiency of the proposed algorithms, comparison by more examples is required. Therefore the author uses more examples with common size. Details of Examples 4-9 has been shown in Table 8 and a comparison among TDM1, TDSM with North-West and Vogel algorithms has been presented in Table 9. Only costs of the Example 4-9 are generated randomly, code of this case has been shown in the last column in Table 8.

Table 6 – Details of Examples 4-9

	Size	S	D	C	Code
Example 4	3×4	[7 9 18]	[5 8 7 14]	41 46 14 49 46 32 28 8 7 5 48 49	randi(50,3,4)
Example 5	5×5	[2 3 5 7 9]	[1 4 6 8 7]	82 10 16 15 66 91 28 98 43 4 13 55 96 92 85 92 96 49 80 94 64 97 81 96 68	randi(100,5,5)
Example 6	10×7	[6 8 11 13 23 5 20 13 14 16]	[14 24 28 33 4 12 14]	17 4 14 15 9 6 16 19 20 1 1 8 14 6 3 20 17 6 16 14 11 19 10 19 1 16 4 14 13 17 14 2 4 3 18 2 3 16 17 10 10 20 6 9 15 14 9 20 11 11 19 8 7 13 7 3 20 16 14 20 15 12 3 20 20 4 1 16 5 6	randi(20,10,7)
Example 7	8×9	[16 18 10 13 11 15 22 23]	[13 21 15 8 10 12 20 13 16]	25 29 13 21 9 14 22 29 27 28 29 28 23 2 12 23 11 29 4 5 24 23 3 23 9 18 17 28 30 29 12 25 24 21 7 5 19 29 20 20 21 6 20 23 5 3 15 2 6 10 15 5 8 8 9 25 26 22 29 14 4 16 26 17 5 29 1 2 20 15 21 8	randi(30,8,9)
Example 8	20×6	[5 18 7 11 20 15 10 11 4 6 21 8 11 10 20 3 7 12 19 21]	[45 56 25 40 41 32]		randi(40,20,6)
Example 9	20×20	[10 12 61 71 31 13 43 92 12 7 34 24 16 13 21 41 27 38 41 21]	[31 71 51 12 10 23 12 82 43 17 24 34 10 19 31 31 47 18 51 11]		randi(40,20,20)

Table 7 – Comparison among TDM1, TDSM and other algorithms

	North-West	Vogel	TDM1	TDSM
Example 4	1451	490	490	490
Example 5	1853	1401	1401	1661
Example 6	1651	740	712	712
Example 7	2251	844	935	979
Example 8	4088	1400	1379	1751
Example 9	12949	4637	3843	3870

Improvement amount of the best algorithm in references by TDM1 has been shown in Table 6. It is easy to see that TDM1 is better than even the best algorithm according to the results in Table 10.

Table 8 – Improvement amount of Vogel algorithm by TDM1

	Vogel	TDM1	Improvement
Example 4	490	490	0%
Example 5	1401	1401	0%
Example 6	740	712	0.037%
Example 7	844	935	-0.1%
Example 8	1400	1379	0.015%
Example 9	4637	3843	0.17%

Using proposed MATLAB code of Vogel algorithm in this paper, solving of large size transportation problem is possible. In this section more large size Examples generated randomly for two reasons: firstly, to compare proposed algorithms here and other methods in references. Secondly to show feasibility of the proposed algorithms and MATLAB code of Vogel algorithm for solving much larger problems. Pseudo code of generation of the problems has been shown in Table 11 and a comparison among TDM1, TDM2 with other algorithms has been proposed in Table 12. All of required information for a transportation problems are generated randomly for the Example 10-15.

Table 9 - Pseudo code of generation transportation problems in large size

<pre> Input(m,n,k);s=randi(100,1,m);d=zeros(1,n); u=1; for i=1:n for j=u: u+1 d(i)=d(i)+s(j); end u=u+2; end c=randi(k,m,n); </pre>

Table 10 – Comparison among TDM1, TDSM and other algorithms for large size problems

	Size	North-West	Vogel	TDM1	TDM2
Example 10	200 × 50	297629	26566	25795	32663
Example 11	160 × 40	185366	85456	66948	13459
Example 12	100 × 25	177666	26462	26184	29502
Example 13	80 × 20	132804	30123	23412	27336
Example 14	210 × 70	322356	27619	27209	28765
Example 15	261 × 87	245311	152930	137248	10534

According to Table 13, results of TDM1 is much better than Vogel algorithm for all different random problems (Examples 10-15). TDM2 is better only for Examples 11, 15. Improvement amount of the Vogel algorithm by TDM1 for large size problems has been shown in Table 8.

Table 11 – Improvement amount of Vogel algorithm by TDM1

	Size	Vogel	TDM1	Improvement
Example 10	200 × 50	26566	25795	0.029%
Example 11	160 × 40	85456	66948	0.216%
Example 12	100 × 25	26462	26184	0.010%
Example 13	80 × 20	30123	23412	0.222%
Example 14	210 × 70	27619	27209	0.014%
Example 15	261 × 87	152930	137248	0.102%

4 Conclusion and future works

In this paper, an efficient algorithm has been proposed which finds better initial basic feasible solution to start simplex method of transportation problems. Lower computational complexity is because of less calculations in the algorithm, in the best case (TDM1), in fact only rows of transportation table are allocated by penalties in this case. Therefore, perhaps this method will be interested by future works in real transportation problems. The proposed algorithm, in all three cases, and some of previous algorithms used to solve several test and generated problems in different sizes. These generated problems and comparisons and also Matlab codes of the best case of the algorithm and Vogel algorithm, can be useful in the future research in transportation area. Finally, a new approach, better than simplex, which can propose optimal solution will be main challenge of the transportation problems. Modification and improvement of proposed heuristic method in this paper, is helpful idea to propose an efficient algorithm to find optimal solution.

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Received: June 3, 2017; Published: July 10, 2017